

# Project Work in Optimization Techniques for Machine Learning: An Empirical Analysis of Optimization Algorithms on Heavily Unbalanced Classification Tasks

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## Abstract

Stochastic Gradient Descent (SGD), when properly tuned, is generally capable of training many neural network architectures at least as well as adaptive methods like Adam. However, a significant performance gap emerges in language modeling tasks, where Adam appears necessary for successful training. Recent literature hypothesizes that this discrepancy is heavily influenced by the extreme class unbalance inherent to these problems. This report investigates this hypothesis by isolating the complexities of this setting through convex, heavily unbalanced two-class problems [1]. We evaluate various first-order and adaptive optimization methods in both full-batch and minibatch regimes. The empirical results confirm that standard SGD methods collapse towards the majority class under severe imbalance, whereas adaptive algorithms like Adam dynamically protect updates for rare features, maintaining significantly higher accuracy on minority classes.

## 1 Introduction

While Stochastic Gradient Descent (SGD) remains a fundamental workhorse in deep learning, its relative performance compared to adaptive methods like Adam varies significantly across domains. In particular, language modeling tasks exhibit a regime where Adam drastically outperforms SGD. A recent study by Kunstner et al. (2024) [1] suggests that this gap is deeply tied to the heavy-tailed class imbalance present in such tasks. Furthermore, it has been hypothesized that the combination of this specific setup with minibatching strategies is the root cause of the performance divergence.

To deepen this analysis and isolate the algorithmic behavior, we test these dynamics in a simplified, convex setting: logistic regression with highly unbalanced binary data. By systematically varying the dataset size and the minority class ratio, we aim to measure the robustness of various optimization algorithms in both Full Batch (FB) and Minibatch (MB) settings.

## 2 Theoretical Background

### 2.1 The Objective Function

The problem is framed as a binary classification task trained via Logistic Regression. The objective is to minimize the Binary Cross-Entropy loss:

$$L(w) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\sigma(x_i^T w)) + (1 - y_i) \log(1 - \sigma(x_i^T w))] \quad (1)$$

where  $\sigma(z) = 1/(1 + e^{-z})$  is the sigmoid function.

## 2.2 Optimization Algorithms

We implemented and evaluated a suite of optimization algorithms:

- **Standard Methods:** Gradient Descent (FB) and Conjugate Gradient (FB), both utilizing Armijo Line Search to guarantee sufficient decrease.
- **Stochastic Methods:** Stochastic Gradient Descent with Polyak stepsize (MB) [3], which computes adaptive step sizes based on theoretical lower bounds.
- **Adaptive Methods:** Adagrad, RMSProp, AdaDelta and Adam, evaluated in both FB and MB regimes. These methods scale the learning rate of individual parameters based on historical gradient norms.

## 3 Methodology

### 3.1 Dataset Generation

A set of binary classification problems was generated using the `make_classification` module from `scikit-learn` [4]. The feature space was fixed at  $n = 16$ . To simulate the heavy-tailed unbalance, experiments were conducted across a grid of sample sizes  $N \in \{500, 2000, 5000\}$  and positive sample fractions  $p \in \{0.1, 0.01, 0.005\}$ .

### 3.2 Evaluation Protocol

For each configuration, algorithms were executed for 100 epochs. We tracked the global training loss and the separated losses with respect to the two distinct classes. Model accuracy was reported independently for the majority and minority classes at the final solutions. As a baseline for the global minimum, the L-BFGS-B method from the `scipy.optimize` package was applied to the full dataset.

## 4 Results and Discussion

### 4.1 The "Accuracy Paradox" and Global Metrics

The empirical results immediately demonstrate that global accuracy is a highly misleading metric under extreme class imbalance. As  $p \rightarrow 0.005$ , global accuracy artificially inflates to  $> 99\%$ . However, observation of separated metrics reveals this is achieved by collapsing into a "predict majority" state (Majority Accuracy  $\approx 100\%$ , Minority Accuracy  $\approx 0\%$ ). Therefore, the true efficacy of an algorithm in this setting is its ability to minimize the minority class loss.

### 4.2 Standard Methods (GD / CG / SGD) vs. The Minority Class

Algorithms applying a uniform step size across all parameters struggled severely. In the Full Batch regime, Gradient Descent and Conjugate Gradient successfully matched the L-BFGS-B global minimum (e.g., Loss = 0.052 at  $N = 5000, p = 0.005$ ); however, the minority class loss spiked to over 3.36, yielding only 3.64% minority accuracy. In the Minibatch regime, SGD with Polyak stepsize exacerbated this failure, as minibatches are overwhelmingly dominated by majority samples, completely washing out minority gradients.

### 4.3 The Power of Adaptive Methods

The results validate the core hypothesis regarding Adam and heavy unbalance. Adaptive methods (Adam, RMSProp, Adagrad) scaled learning rates individually based on historical gradients.

In the Minibatch regime, features associated with rare minority samples generated gradients infrequently. Consequently, their historical squared gradient sums remained low, leading to a much larger effective learning rate when those features did appear. At  $N = 5000, p = 0.01$ , while SGD achieved only 5.00% minority accuracy, Adam reached 23.75% and RMSProp 15.00%, successfully resisting the majority class collapse.

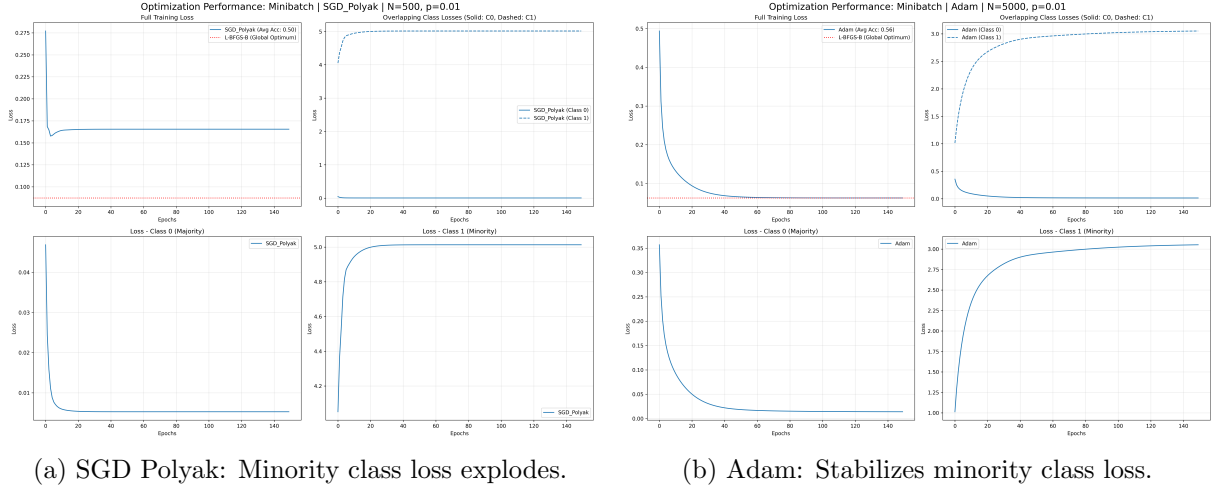


Figure 1: Comparison of SGD and Adam in the Minibatch regime under heavy unbalance ( $N = 5000, p = 0.01$ ). Notice how Adam protects the minority class updates, whereas SGD completely collapses towards the majority class.

#### 4.4 Regime Dynamics: Full Batch vs. Minibatch

An interesting divergence was observed: adaptive methods often failed to reach the global minimum loss in the Full Batch regime (e.g., Adam reaching 0.355 vs L-BFGS-B's 0.049 at  $N = 5000, p = 0.005$ ). However, when switched to the Minibatch regime, their final global losses dropped, aligning almost perfectly with the L-BFGS-B baseline. This confirms that the combination of stochasticity (minibatching) and adaptive scaling is required to successfully navigate the unbalanced topology.

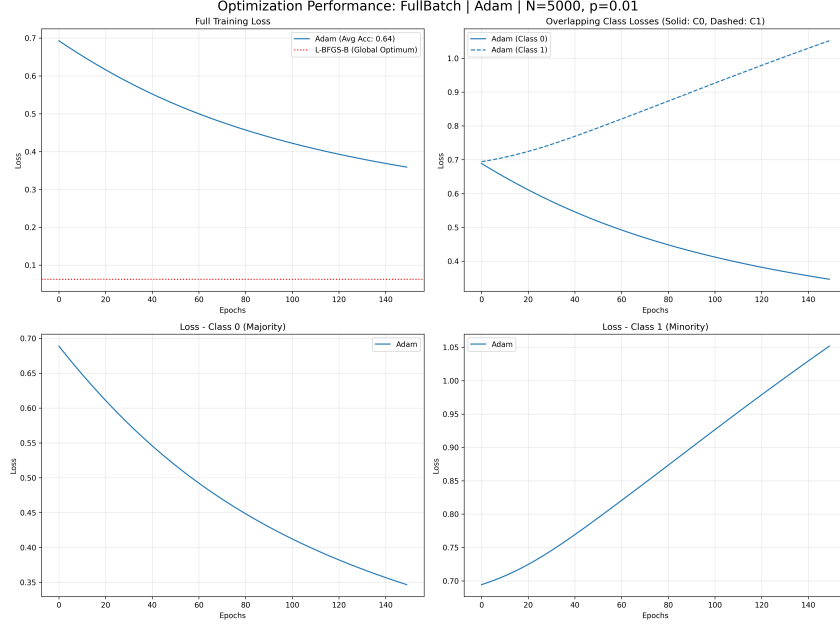


Figure 2: Adam executed in Full Batch regime ( $N = 5000, p = 0.01$ ). Compared to its Mini-batch counterpart (Fig. 1b), the algorithm struggles to smoothly converge, demonstrating that adaptive scaling relies heavily on the stochasticity of minibatches.

#### 4.5 The Absolute Limit of Data

Finally, the experiments demonstrated theoretical limits based on sample size. At  $N = 500, p = 0.005$ , the dataset contains a mere 8 positive samples. In this extreme scenario, all minibatch algorithms, including Adam, yielded near 0.00% minority accuracy. Optimization techniques cannot overcome an absolute lack of statistical representation to form a reliable decision boundary.

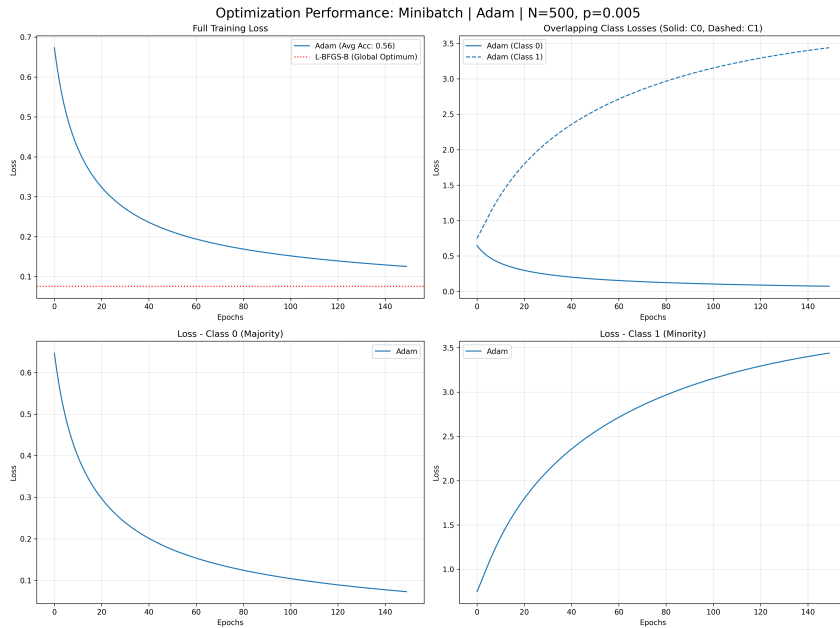


Figure 3: Adam performance under extreme data starvation ( $N = 500, p = 0.005$ , containing only 8 minority samples). The algorithm fails to establish a reliable decision boundary, leading to an exploding Class 1 loss despite adaptive learning rates.

## 5 Conclusion

This empirical study confirms the hypothesis that heavy class imbalance is a primary driver of the performance gap between SGD and Adam in tasks analogous to language modeling. Standard methods are easily overwhelmed by majority class gradients. In contrast, Adam’s parameter-specific learning rates allow it to extract and preserve meaningful updates from rare minority samples, provided that a minibatching strategy is employed to leverage the stochastic nature of the updates.

Table 1: Final performance metrics for selected optimization algorithms across key datasets. This table highlights the critical differences between standard and adaptive methods, particularly regarding minority class accuracy under heavy unbalance.

| Dataset      |               | Method (Regime) | Accuracy (%) |               | Final Loss   |               |
|--------------|---------------|-----------------|--------------|---------------|--------------|---------------|
| Size ( $N$ ) | Ratio ( $p$ ) |                 | Global       | Minority (C1) | Global       | Minority (C1) |
| <b>5000</b>  | <b>0.01</b>   | GD Armijo (FB)  | 99.09        | 11.25         | 0.065        | 2.885         |
|              |               | Adam (FB)       | 93.93        | 32.50         | 0.359        | 1.052         |
|              |               | SGD Polyak (MB) | 97.44        | 5.00          | 0.132        | 3.878         |
|              |               | Adam (MB)       | <b>99.10</b> | <b>12.50</b>  | <b>0.063</b> | <b>3.052</b>  |
| <b>5000</b>  | <b>0.005</b>  | GD Armijo (FB)  | 99.52        | 3.64          | 0.052        | 3.361         |
|              |               | Adam (FB)       | 94.19        | 29.09         | 0.356        | 1.099         |
|              |               | SGD Polyak (MB) | 99.52        | 3.64          | 0.130        | 6.333         |
|              |               | Adam (MB)       | <b>99.53</b> | <b>5.45</b>   | <b>0.049</b> | <b>3.589</b>  |
| <b>500</b>   | <b>0.005</b>  | GD Armijo (FB)  | 99.50        | 0.00          | 0.078        | 3.605         |
|              |               | SGD Polyak (MB) | 97.88        | 0.00          | 0.174        | 4.165         |
|              |               | Adam (MB)       | <b>98.75</b> | <b>12.50</b>  | <b>0.125</b> | <b>3.439</b>  |

## References

- [1] F. Kunstner et al., *"Heavy-tailed class imbalance and why Adam outperforms gradient descent on language models."* Advances in Neural Information Processing Systems (2024).
- [2] M. Lapucci (2025). Course Lecture Notes.
- [3] N. Loizou et al. *"Stochastic Polyak step-size for SGD: An adaptive learning rate for fast convergence."* International conference on artificial intelligence and statistics PMLR (2021).
- [4] scikit-learn library: <https://scikit-learn.org/stable/>